Deep Learning

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Lecture 1: Introduction

1.1 Learning representations

- Machine learning fits an input-to-output function from data when writing its rules by hand is difficult. Its parameterization determines the functions it can express.
- Features describe inputs in a form useful for a prediction. A shallow approach may use hand-designed features and learn a simple classifier on top. Choosing good features can be difficult.
- Representation learning learns features from data. Deep learning uses multiple layers of learned representations, usually implemented by a neural network.

For input x, a model composes parameterized transformations:

$$h_0 = x, \qquad h_i = f_i(h_{i-1}; \theta_i), \qquad \widehat{y} = f_L(h_{L-1}; \theta_L).$$

Here h_i is a representation, θ_i is a layer's parameters, and $\widehat{y}$ is the output. Later representations can become more abstract and less sensitive to irrelevant variation. In end-to-end learning, parameters across the model are trained against a shared task objective; not every deep model is trained this way.

1.2 Example: multilingual translation

A translation model can learn from paired data: sentences expressing the same content in two languages. A separate model for each pair needs suitable data for each pair, which is difficult for rare languages. A multilingual model shares parameters among pairs and is told the desired target language.

Example: Suppose the training directions are English → French, French → English, and English → Spanish. Translating French → Spanish without training examples for that direction is zero-shot translation. Sharing can help transfer information between pairs, but successful transfer is not guaranteed.

The model's internal representation may capture structure useful across languages. Interpolating between target-language signals can produce mixtures; this does not prove that the model has a unique human-like representation of meaning.

1.3 Capacity, bias, and scale

- Model capacity: informally, the range of functions a model class can represent. For example, nonlinear decision boundaries offer possibilities beyond linear boundaries.
- Inductive bias: assumptions built into a learning system that favor some solutions over others. Learned features reduce manual feature design, but architecture and training still introduce biases.
- Scaling: a method can improve as data, model capacity, and computation increase. The lecture identifies large models, datasets, and compute as contributors to deep learning's progress; they also have costs and do not guarantee improvement.

The history runs from the perceptron and backpropagation through handwriting recognition with LeNet to AlexNet's 2012 ImageNet result. Neural networks take their name from a rough analogy to biological neurons; artificial units are simplified computational models.

Source. Sergey Levine, Deep Learning, Spring 2021, Lecture 1.